Driver Drowsiness Detection & Alarm System

DriveSafe

Capstone Project Report

MID SEMESTER EVALUATION

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ABSTRACT

This project aims to build a smart system that monitors the driver's actions and gestures and alerts them of any issue at the appropriate time. It will be completely self-contained and will direct the driver. In the future, this will significantly minimize collisions and fatalities. Furthermore, if the driver cannot undertake activities on their own, automatic braking systems will be used. As a result, the growth of the country's municipal infrastructure will be aided by such an integrated system. The recommended method is straightforward but effective in addressing our country's road safety crisis. The safety system will constantly monitor the driver, which will issue warnings that will become increasingly severe if no action is done. The use of current sensors placed in automobiles and their integration with the camera is an intriguing essential feature of this method. Because the solution relies primarily on video surveillance, the cameras must be placed directly in front of the driver, out of sight. Installing cameras and connecting them with sensors will be cost-effective, efficient, and dependable because most of the hardware is already in place.

ACKNOWLEDGEMENT

We would like to express our thanks to our mentor Ms Swati Kumari . She has been of great help in our venture, and an indispensable resource of technical knowledge. She is truly an amazing mentor to have.

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They always wanted the best for us and we admire their determination and sacrifice.

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DECLARATION

We hereby declare that the design principles and working prototype model of the project entitled Driver drowsiness detection and Alert system is an authentic record of our own work carried out in the Computer Science and Engineering Department, TIET, Patiala, under the guidance of Ms. Swati Kumari during 7th semester (2022).

Date: 26 -08-2022

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INTRODUCTION

1.1 Project Overview

National Highway Traffic Safety Administration (NHTSA) research information demonstrates that languor while driving is a contributing component for street mishaps and it results in 4-6 times higher accident hazard with respect to caution drivers. A large portion of the lethal street mishaps happen at rates more noteworthy than 70 Kmph. The World Health Organization (WHO) has revealed that India has the most noticeably bad street conditions on the planet coming about roughly over two lakh passing in 2010 and 2011.

Research demonstrates that driver weariness and drowsiness is one of the real purposes behind the expanding mishaps. India is one of the busiest countries globally in terms of road traffic. Road accidents are recently increasing rapidly and have become a concern for the government and regulatory boards. India ranks first in the number of road accident deaths across the 199 countries and accounts for almost 11% of the accident-related deaths in the World. In 2021, India's Union Minister Mr. Nitin Gadkari cited 'Road accidents are more dangerous than the COVID-19 pandemic,' reporting 1.50 Lakh of life compared to 1.46 Lakh because of COVID-19. his translates to a loss of Rs 91.16 Lakh per person lost, which accounts for 3.14 percent of GDP annually.

Driver exhaustion not just affects the readiness and reaction time of the driver yet it additionally improves the odds of being engaged with fender benders. The lethargic drivers neglect to take right activities before an impact. An essential incongruity in driver's weakness is that the driver might be too depleted to even consider comprehending his own dimension of laziness. This critical issue is regularly disregarded by the driver. Thus, the utilization of supporting frameworks that look at a driver's dimension of watchfulness is important to maintain a strategic distance from street mishaps. These frameworks should then caution the driver on account of lethargy or obliviousness. Some notice signs that can be estimated as signs of driver weakness seem to be: wandering off in fantasy land while out and about, rolling over the inside line, yawning, feeling fretful, feeling hardened, overwhelming eyes, and responding gradually.

There are a lot of factors behind these fatal crashes, but the major contributors are driving under drug/alcohol consumption, use of mobile phones, and driver fatigue. About 70% of the victims are India's young population between 18 and 45, which is alarming.

Engineers and Scientists are investing billions of dollars in self-driving vehicles to minimize road crashes but, this technology has to refine itself, which will take more money and time. The effort to upgrade and improve the existing technology is little or none. We propose a solution to this growing problem by inculcating modern object detection techniques so that the present obsolete system can be replaced with a much faster, better, and smarter system that is responsive to traffic conditions and adjusts accordingly. The proposed new system, once implemented, will be fully automatic and guide the driver along the way. The main aim of this project is to make a smart system that monitors the driver's actions, and gestures and makes the driver aware of the situation at the right time. This will considerably reduce crashes and loss of lives in the coming time. In addition to this, if the driver cannot perform actions by themselves, then automatic braking systems will come into the action. Thus, such an integrated system will benefit the civic development of the country.

The proposed solution is simple yet effective in solving the road accident crisis of our country. The safety system will continuously monitor the driver and give warnings that will get intense if no action is taken. The interesting key point of this solution is utilizing the existing sensors installed in the vehicles and integrating them with the camera. Since the solution is heavily based upon monitoring using cameras, there is an utmost need for the cameras to be set up in front of the driver, away from the driver's sight. Since most of the hardware is already installed in vehicles, installing cameras and integrating them with sensors will be economical, efficient, and reliable.

1.1.1 Tecnologies

- Arduino
- Open Cv
- Wireless Modules

1.1.2 Problem Statement

The main aim of this project is to make a smart safety system that detects all gestures and facial landmarks of the driver continuously give the warning at regular intervals, and finally applies automatic brakes to save the driver from the crash. There is an ever-increasing number of crashes which has become a serious life threat. The vehicle accident prevention system will use an eye blink sensor for alertness monitoring and drowsiness detection. Driver face monitoring systems capture the images from the driver's face and extract the symptoms of fatigue and distraction from the eyes, mouth, and head. These symptoms are usually a percentage of eyelid closure over time (PERCLOS), eyelid distance, eye blink rate, blink speed, gaze direction, eye saccadic movement, yawning, head nodding, and head orientation. The system estimates driver alertness based on extracted symptoms and alarms if needed

1.1.3 Objective

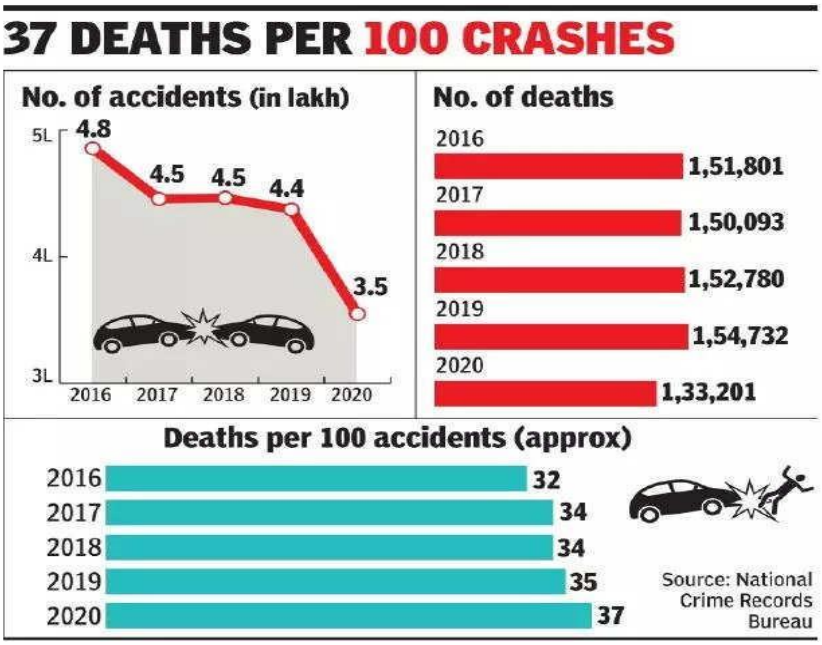
This project aims to develop an advanced automobile safety system based on the impending widespread availability of probe data of facial gestures, leading to a transformational change in how we detect drowsiness and distraction and then use this knowledge to make road journeys safer. The vision of realtime information to transform the driving experience offer the potential for significant improvements in the efficiency of the vehicle safety system. Ideally, knowledge of the location, speed, road type, and neighbor traffic of every vehicle in the network in real-time allows for optimal braking management. However, traditional infrastructure-based traffic detectors only provide point-specific information on the presence of vehicles. With this knowledge, meaningful performance measures can be estimated accurately for the driver

1.1.4 Proposed Solution

This project aims to develop an advanced automobile safety system based on the impending widespread availability of probe data of facial gestures, leading to a transformational change in how we detect drowsiness and distraction and then use this knowledge to make road journeys safer. The vision of realtime information to transform the driving experience offers the potential for significant improvements in the efficiency of the vehicle safety system. Ideally, knowledge of the location, speed, road type, and neighbor traffic of every vehicle in the network in real-time allows for optimal braking management. However, traditional infrastructure-based traffic detectors only provide point-specific information on the presence of vehicles. With this knowledge, meaningful performance measures can be estimated accurately for the driver.

1.2 Need Analysis

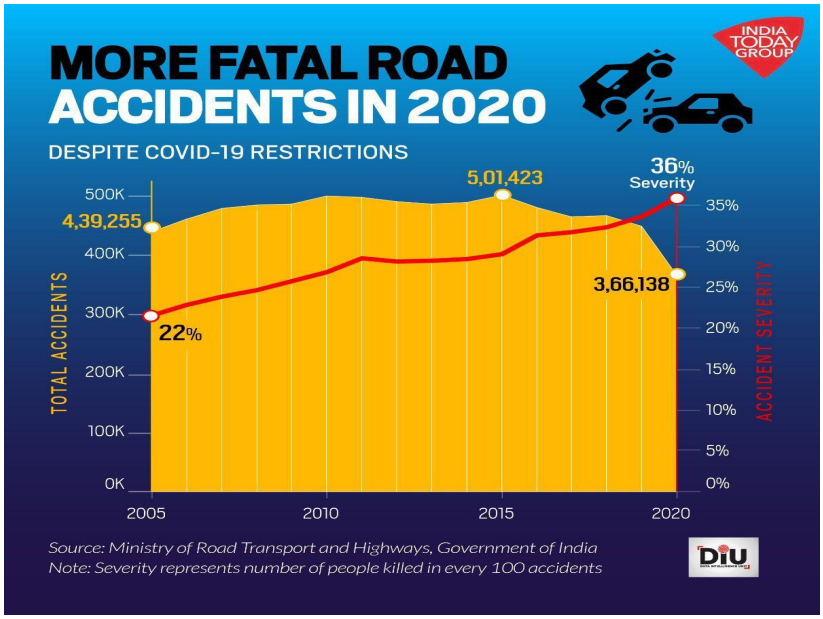
Figure 1: 37 Deaths per 100 Crashes



A question may arise that in the above image there has been a decline in the number of road accidents, then what is the need of any prevention system. So, the answer to the question lies in the figures given in the image that even after the declination the number of deaths per 100 accidents is still increasing, which clearly

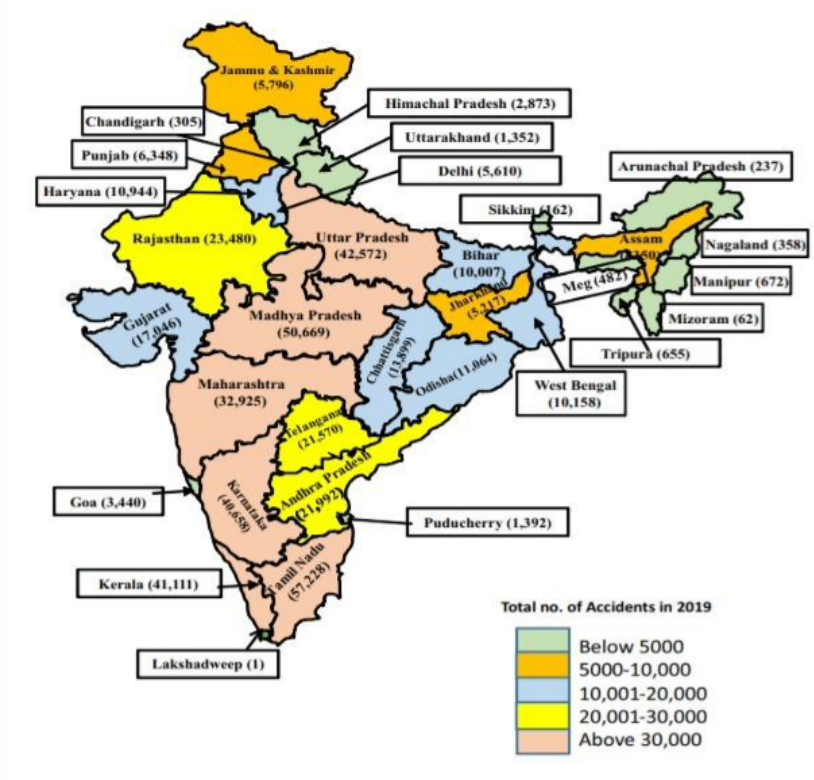
shows that declination is not that significant for which we need some effective road accident prevention system which could make a significant reduction in this figure.

Figure 2: Severity of Road accidents



So, according to various researches, the reason for about 20% of the crashes and 30% of fatal crashes is driver drowsiness and lack of concentration. Fatigue reduces the driver's perception and decision-making capability to control the vehicle. A trend in research shows that after 2-3 hours of continuous driving, the driver is fatigued, and steering performance deteriorates. According to the current studies, it is expected that the number of crashes will be reduced by 10% to 20% using driver face monitoring systems.

FIGURE 3: Statewise number of Accidents in 2019



1.3 Research Gaps

Ref. No.	Authors and Year	Research Gaps
[2]	Aliyu, Ahmed, et al. (2017) 570-576	1. It doesn't detect obstacle detection at two sides of the vehicle. 2. It doesn't consider relative velocity between vehicle and obstacle as obstacle detected can also be moving state.
[3]	Vidhya, D. S., et al. (2016): 316-320	1. It doesn't alert the driver if an obstacle is detected. 2. In case of an obstacle moving towards the vehicle, the brakes are applied much fast so the immediate impact of brakes can injure the driver.
[4]	Deng, Wanghua, and Ruoxue Wu. (2019): 118727-118738.	1. They didn't consider the vertical head movement, which can also be important to detect if a person has dozed off or not. 2. No emphasis was given to alert the driver on detection of drowsiness
[5]	Abraham, Steffin, Tana Luciya Joji, and D. Yuvaraj. (2018): 2295-2310.	1. They didn't consider the vertical head movement, which can also be important to detect if a person has dozed off or not. 2. No emphasis was given to alert the driver on detection of drowsiness
[6]	Alhassan, Mohammed & Eduku, Stephen & Sarpong, Godfred. (2016). Volume 2	1. Eye Blinking sensor used, which has less accuracy.
[7]	Eduku, Stephen & Alhassan, Mohammed & Sekyi-Ansah, Joseph. (2017)	1. The IR sensor used here is less accurate than the computer vision used in this project. 2. No emphasis is given to driver safety if he falls asleep

Table 1: Research Gaps

1.4 Problem Definition and Scope

If observed the reason behind many day-to-day accidents, it would be either because of feeling drowsy or being distracted and looking at the number of such accidents one may feel that this is one of the major problems prevailing in the country which demands a promising and reliable solution which could reduce these figures remarkably and can save a lot of lives. Therefore, we will be working upon this issue and come up with the best possible solution.

This project is being done with a future vision. Numerous prevention techniques are being adopted and implemented to ensure one's life and our society and automobile market too is very open to new techniques or ideas that can ensure safety to the best possible level. So, our idea is one of the promising solutions and is also very likely to get adopted by society and the market since not much overhead is currently present, and it will also be affordable for the common public.

1.5 ASSUMPTIONS AND CONSTRAINTS

Assumptions

1. Continuous power should be supplied to all the components.
2. Some percentage of eyes is open is already considered
3. The camera lens should work properly and must not be malfunctioned

Constraints

1. Good light conditions are necessary. The results may differ in low light conditions.
2. Results with accessories like spectacles, masks etc may be different and less accurate.

1.6 STANDARDS

Code Standard

- Our code is conformed to PEP 8, the Style Guide for Python Code.
- The code layout and white-space, comments, imports, documentation strings, version control, programming recommendations, and naming conventions are conformed to PEP 8.
- We have also used a few deviations from the Basic Standard.

Safety Standards

- ISO 13849 safety standard has been followed.

- It deals with machinery safety and safety-related design principles of control systems.

Software Process Quality Standard

- ISO/IEC 15504, also known as Software Process Improvement and Capability Determination (SPICE), is a technical standards documents for the computer software development process.
- This standard is gone for setting a clear model for process correlation. It shows procedures to oversee, control, guide, and screen programming improvement. This model is then used to gauge what a task group does amid programming improvement. This data is dissected to recognize shortcomings and drive enhancement.

Other Standards include:

- ISO20954 for image stabilization and the camera.
- ISO19090- low light performance.
- ISO20490- autofocus standard for the camera.

1.7 APPROVED OBJECTIVES

The negligence by the drivers costs an average of 1214 crashes in India daily. To counter the problem, the Government of India is spending a whopping 20,000 crores a year but sadly that does not seem to affect the number of accidents taking place. To find a solution to this complex problem, we are working towards implementing a Car Safety System which is an economical endeavor to solve this persistent issue.

Our goal is to track the driver's movements during the journey and make appropriate decisions to alert the driver before the consequences. To implement this safety system, we will be using facial points to track the driver's movements. To improve the system's accuracy, it is important to track multiple points across the face and multiple sense organs like the eyes, mouth, nose and ears for a single feature.

We will be using chin and nose tracking to check the driver's line of sight. For checking drowsiness, we will be tracking neck, chin, and eye angles. If in the case where alarming seems to put no effect on the driver, the system will try to bring the moving automobile into retardation as a prevention system by performing calculations about the distance of the previous and next vehicles. This proposal focuses on providing a safe ride for the driver.

The overall objective of the project is the following:

- Alarming driver in case of distraction.
- Informing driver in case of any drowsiness detected.

- Using automatic brakes upon obstacle detection.

1.8 METHODOLOGY

1. We start with a robocar (buggy) having an Arduino microcontroller.
2. Arduino is used to control the movement of buggy, its 4 ultrasonic sensors and Bluetooth wireless module for input from the laptop.
3. As the project is only a prototype, to record, a real-time video laptop camera is used.
4. The real-time video thus recorded is evaluated to detect facial landmarks of drivers and further detect drowsiness.
5. On detecting drowsiness, a signal is sent from laptop's Bluetooth to buggy's, triggering an alarm system.
6. If the driver is alert, the alarm is stopped, and if the driver doesn't respond, a braking sequence is initiated based upon the current speed of the car and the distance of adjacent cars on the lane.

1.9 PROJECT OUTCOMES AND DELIVERABLES

- A prototype robocar will be ready to display every functionality of our project.
- The model will detect driver’s drowsiness.
- Various reactions will be taken by our model depending upon the actions
    i. Drowsiness: Alert system will be activated and the vehicle will begin to slow down.
    ii. Obstacle Detection and Alarm System: When the driver is drowsy the sensors get activated and a buffer timer starts. If the driver is unable to be alert again within the buffer time the braking system is gradually applied.

1.10 NOVELTY OF WORK

Our main goal is to prevent road accidents that occur due to drowsiness, for they are one of the major reasons behind accidents in day-to-day life. Though many Preventions Systems or techniques have been proposed to ensure safety, we tried to cover many gaps in our idea. We incorporated them in our idea, making it way different and efficient than any other proposed solution. So basically, our solution is mostly a bag of borrowings that fulfils research gaps given out by other proposed ones.

Earlier, the focus was on either prevention or detection, but we will club both of them to warn before activating the prevention system. Then if we talk about the detection system which was earlier proposed, their main task was just to detect. In contrast, we will incorporate various alerting mechanisms in our idea to warn drivers first. Many researches considered very few parameters to detect, resulting in many gaps and ensuring less efficiency, which we will try to cover since we will consider various landmarks to be very

precise and ensure the best possible efficiency. Moreover, many used some sensors in detection, whereas we will be detecting using the OpenCV technique to gain more accuracy.

Then coming to prevention system, first of many fewer researches talk about prevention and which do, many of them had a braking system, but that was independent of vehicle's speed and without any gradual retardation which can cause injury to the driver due to sudden brakes which we will take care and will try to implement automatic braking system ensuring gradual retardation. Many of the proposed solutions do not warn the driver if some sort of error is there or activation of a prevention system is not required. In contrast, we do keep buffer time in which we will just activate an alert mechanism to warn him first and alongside sensors to detect any obstacle.

So, as a whole, we will try our best to cover most of the gaps or limitations in the earlier proposed solution and many of our ideas, which will make it unique and ensure Novelty of work.

2.1 Literature Survey

2.1.1 Theory Associated with Problem Area

Driver Fatigue is among the most widely recognized explanations behind deadly street mishaps around the globe. This demonstrates in the transportation business particularly, where a driver of a substantial vehicle is regularly presented to long stretches of tedious driving which causes weariness without a continuous rest period. Because of the successive event of driver weariness this has turned into a territory of incredible financial concern. Recognizing driver's sleepiness has been an exploration point for a long time, with numerous methodologies being considered along.

When one considers the number of accidents that occur, it is clear that this is one of the country's key challenges, one that necessitates a viable and trustworthy solution that may drastically lower these figures and save not all but the majority of lives. Driver tiredness and lack of focus, according to several studies, is the cause of roughly 20% of crashes and 30% of fatal crashes. According to current research, driver facial monitoring devices can minimize crashes by 10% to 20%.

2.1.2 Exixting system and solutions

Several manufacturers, including Audi, Mercedes and Volvo, currently offer drowsiness detection systems that monitor a vehicle’s movements, such as steering wheel angle, lane deviation, time driven and road conditions. When drowsiness is detected, drivers are typically warned with a sound and the appearance of a coffee cup icon.

Plessey Semiconductors has developed sensors, to be placed in a seat, that monitor changes in heart rate. Algorithms developed by the company indicate when breathing changes to patterns that are typical of someone who is sleeping, giving a warning before someone actually feels tired. Some have developed the systems which monitor the driver’s point of attention

2.1.3 Research findings for existing Literature

Table 2: Research findings for existing Literature

S.No	Roll Number	Name	Tools/ Technologies	Findings	Citation
1	101903569	Sai Sharan Sakilam	Object Detection using Ultrasonic sensor HC-SR04	On detecting obstacles in front or back of the car, the braking sequence was applied relative to the speed of the vehicle and obstacle.	[1]

2	101903569	Sai Sharan Sakilam	Automatic Braking using Ultrasonic senor	The time between sending and receiving the signals is used to check the relative distance between vehicle and obstacle.	[3]
3	101903569	Sai Sharan Sakilam	Accident Prevention System Using Blink Sensor	It alerts the driver and initiates adaptive brakes on detecting drowsiness using an eye blink sensor if some threshold time limit is crossed.	[13]
4	101903172	Jagjeevan Singh	Drowsiness detection using facial features	The eyes and mouth facial contours are used to check for driver drowsiness	[14]
5	101903172	Jagjeevan Singh	Drowsiness detection using Machine Learning	Machine learning algorithms and computer vision can identify driver fatigue in a real-time video stream and inform the driver by playing an alarm if they look asleep.	[8]
6	101903682	Saksham Chauhan	Driver Drowsiness Detection using Artificial Intelligence	It detects the driver's state via Face detection, Eye position detection, and Eye-tracking. This system detects the driver's status based on eye movement and provides an alert in less than 0.5 seconds.	[9]
7	101903682	Saksham Chauhan	Accident Prevention System Using IR Sensor	This sensor also works as an eye blink sensor, but the detection method is somewhat different as it detects drowsiness using IR transmitters and receivers over a period.	[7]

2.1.4 Problem Identified

Considering the number of daily accidents that occur in the country, it is evident that this is one of the country's biggest concerns, one that demands a feasible and reliable solution capable of dramatically lowering these figures and saving not all but the vast majority of lives. After long hours of driving and continuous fatigue compromises the ability of the driver to make decisions. This also leads to low steering performance and high risk of road mishap.

2.1.5 Survey on Tools and technologies used

- Arduino Microcontroller: Arduino is a free and open-source platform that includes both hardware and software that is simple to use. Arduino is a beginner-friendly platform. Arduino boards can be used to create a variety of projects and applications. The Arduino IDE, which is free software, is used to write the instructions that your microcontroller must follow. One of the most widely used Arduino microcontrollers is the ATMega 328P.
- Face detection: Face detection is a vital first step in face recognition systems that allows the face region to be located and extracted from the background. It can also be used for content-based picture retrieval, video coding, video conferencing, crowd monitoring, and intelligent human computer

interactions. Face detection in computer vision is a difficult problem because the human face is a dynamic object with great diversity in its appearance. A wide range of methodologies has been proposed, ranging from simple edge-based algorithms to composite high-level approaches incorporating powerful pattern recognition methods.

2.2 Software Requirement Specification

2.2.1 Introduction

2.2.1.1 Purpose

The major purpose of this project is to enhance the driver's safety while driving the vehicles by detecting their face for drowsiness or distraction and implementing a prevention and avoidance sequence to avoid any accidents.

2.2.1.2 Intended Audience and Reading Suggestions

This document is intended for the vehicle owners and manufacturers who wish to install the technology for the safety of drivers. This document also serves as a contract between the product owner and the developers where the owner can see what and how the developers intend to make the product. Reading suggestions are different to different readers. So, one should take the appropriate manner to read the document depending on their role.

2.2.1.3 Project Scope

The primary goal is to detect whether the driver is drowsy or distracted and when detected trigger proper alert system, breaking sequence using algorithms with real time data from the sensors.

2.2.2 Overall Description

2.2.2.1 Product Prespective

This product is a powerful yet simple prototype that provides a single platform for efficient detection of drowsiness and distraction. The product requires an Arduino microcontroller and sensors to control the sensors and trigger various warnings to alert the driver and preventive measures to avoid any accidents. The user's live face capturing will be done using the camera. This video, thus captured in real-time, is enhanced so that operations applied on the face can be done more accurately. The operations include doze, yawn, and distraction detections using some calculations on facial landmarks. Further, the result will be stored in the database for future reference. All these functions will be done as fast as possible so that no information between successive video frames is lost. All basic functions are written in the Product Features topic below

2.2.2.2 Product Features

- Real-time driver’s face video capturing
- Real time drowsiness detection,
- Triggering audio alerts/warnings

- Gradual deceleration of vehicle using braking sequence
- Nearby vehicle detection using ultrasonic sensor

2.2.3 External Interface Requirements

2.2.3.1 User Interfaces

This project is hardware simulation-based, so no user interface is involved.

2.2.3.2 Hardware Interfaces

- Arduino Uno is a microcontroller board based on ATmega328P microchip
- Sensors like Camera, Ultrasonic sensor (HC-SR04)
- HC-05 Bluetooth module to control Arduino wirelessly
- Buggy
- Some LEDs and buzzers to show the alert signals

2.2.3.2 Software Interfaces

- Arduino IDE for Arduino programming
- Jupyter notebook as a development tool

2.2.4 Other Non-functional Requirements

2.2.4.1 Performance Requirements

- Response Time: The system should be fast. It should be quick enough to collect the driver's facial expressions, process them and then respond with necessary action within a fraction of seconds.
- Workload: The system should be able to process the data quickly and efficiently even with the increase in workload, as in the case of image enhancements needed in low light and other conditions.
- Reliability: The system must be reliable enough to take necessary actions in any situations that require the driver's attention or in any other emergency condition.
- Availability: The system must be running when the driver is driving the vehicle without downtime or lapses.

2.2.4.2 Safety Requirements

- A backup power supply should be available if the main power supply fails for any reason.
- The system must be weather-proof such that its working is not affected by any kind of harsh weather.
- The system must be insulated from contacts with external energized electrical circuits.
- The sensors must not be covered or dirty else; it could make the system ineffective.

2.2.4.3 Security Requirements

- The hardware must be tamper-proof so that it cannot be tampered with in any case as the important functionalities of the vehicle (like the braking system) are controllable by the system.
- The data stored on the local storage must be encrypted to prevent unauthorized access to the data.

2.3 Cost Analysis

Table 3: Cost Analysis

Sr. No	Hardware Components	Price (In Rupees)
1.	Robocar/ Buggy	4100
2.	Microcontroller AT Mega 328	1250
3.	Alarm/ Buzzer	120
4.	Bluetooth Module/ Wireless communication	400
5.	Ultrasonic Sensors	1100
6.	Connecting Wires and Batteries	400
7.	Miscellaneous	350
	Total	7720

2.4 Risk Analysis

Table 4: Risk Analysis

Sr. No	Risk	Probability	Alleviation
1.	Microcontroller or sensor components fail or become faulty	0.3	Use better quality products and make more strong connections
2.	The input for the face detection may give delayed input.	0.2	Use more complex algorithms song with a better quality dataset.
3.	The project is not completed on time	0.2	Make proper workflow and adhere to the deadlines.

3.1 Investigative Techniques

Table 5: Investigative Techniques

S.No.	Investigative Projects Techniques	Investigative Techniques Description	Investigative Projects Examples
1.	Testing	To find the link between the factors influencing the outcome. Each variable is modified one at a time, while the others remain constant, and the change resulting from each variable is calculated.	It's great for technology inquiries and especially good for measurement-based investigations.
2.	Categorization	It is based on a method of grouping and categorizing occurrences or objects. Keys are frequently employed as criteria in the procedure.	Projects based on segregation and categorization. Projects involving databases and information retrieval are the most common.
3.	Designing	Designing models can help you figure out how and why a process operates. Some models are pre-defined, while others must be developed.	All of the initiatives are based on cutting-edge data analytics and machine learning approaches.
4.	Pattern Following	Observing and recording natural events, or conducting experiments with variables that are difficult to control, are examples of this technique.	Face identification, text classification, caption creation, and object detection are just a few examples of projects that have used this technology.

3.2 Proposed Solution

Accidents are caused by various factors, one of which is negligence. We are attempting to create a cost effective Car Safety System to address this never-ending problem to address this issue. This system will detect any negligence on the driver's part and will quickly notify him. It would also be capable of taking any necessary actions in an emergency.

Our goal is to keep track of the driver's movements during the journey and make necessary judgments to make the driver aware of the situation before it becomes a problem. We'll use facial points to track the driver's motions to build this safety system. Various landmarks on the driver's face must be tracked and combined into a single feature to improve the system's accuracy. DriveSafe will use a camera to track the driver's movements during the journey, and it will be able to take appropriate action, if necessary, after evaluating the driver's facial landmarks. In low-light situations, it would conduct picture enhancements as needed.

Drowsiness detection will be aided by eye-tracking. Suppose the driver's eyes are closed below a threshold limit for a predetermined amount of time. In that case, DriveSafe will start notifying him, either with an alarm or by raising the audio system level to get his attention. Suppose alarming appears not to affect the driver. In that case, the system will attempt to slow down the driving vehicle as a preventative measure by measuring the distance between the preceding and next vehicles.

DriveSafe will try to alert the driver and instruct him to refocus his vision on the road. The project will also maintain track of the driver's activities as a record that he may use to check his journey's statistics.

3.3 Work Breakdown Structure

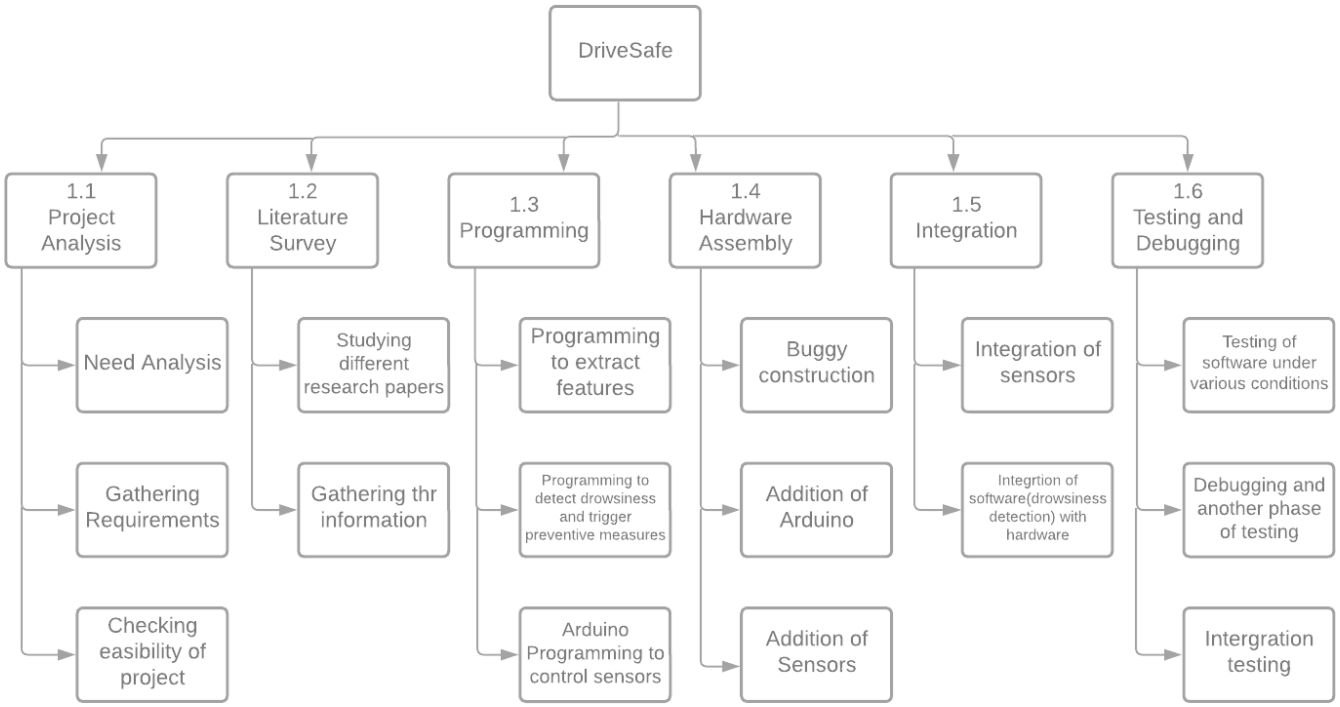


Figure 4: Work Breakdown Structure

In Problem Statement (Section 1.1) of Work Breakdown Structure:

- The need and scope of the project were analysed to make the project more useful and analyse tasks required to accomplish the project.
- All functional and non-functional requirements were taken into account for better project planning.
- A feasibility study was performed to make the project efficient and less tedious by helping in a better decision-making process.

In Literature Survey (Section 1.2) of Work Breakdown Structure: Studying research papers gave us ideas on areas that we can work and build upon to make our project better reliable, and robust.

In Programming (Section 1.3) of Work Breakdown Structure:

- Image enhancement technique is performed for better detection at night and in a hazy atmosphere.

- Facial landmark detection is performed, and other landmarks around eyes, face, mouth movements are analysed to detect drowsiness.
- The algorithm generates various warnings on the type and severity of distraction and drowsiness to alert the driver.

3.4 Tools and Technology

Tools/Platform:

- Python with Computer Vision (OpenCV) and Signal libraries
- Arduino
- Jupyter Notebook

Hardware:

- Buggy
- ATmega328P Microcontroller
- 4 Ultrasonic sensors
- Bluetooth module
- Laptop Webcam
- H-bridge
- Buzzer

4.1 System Architecture

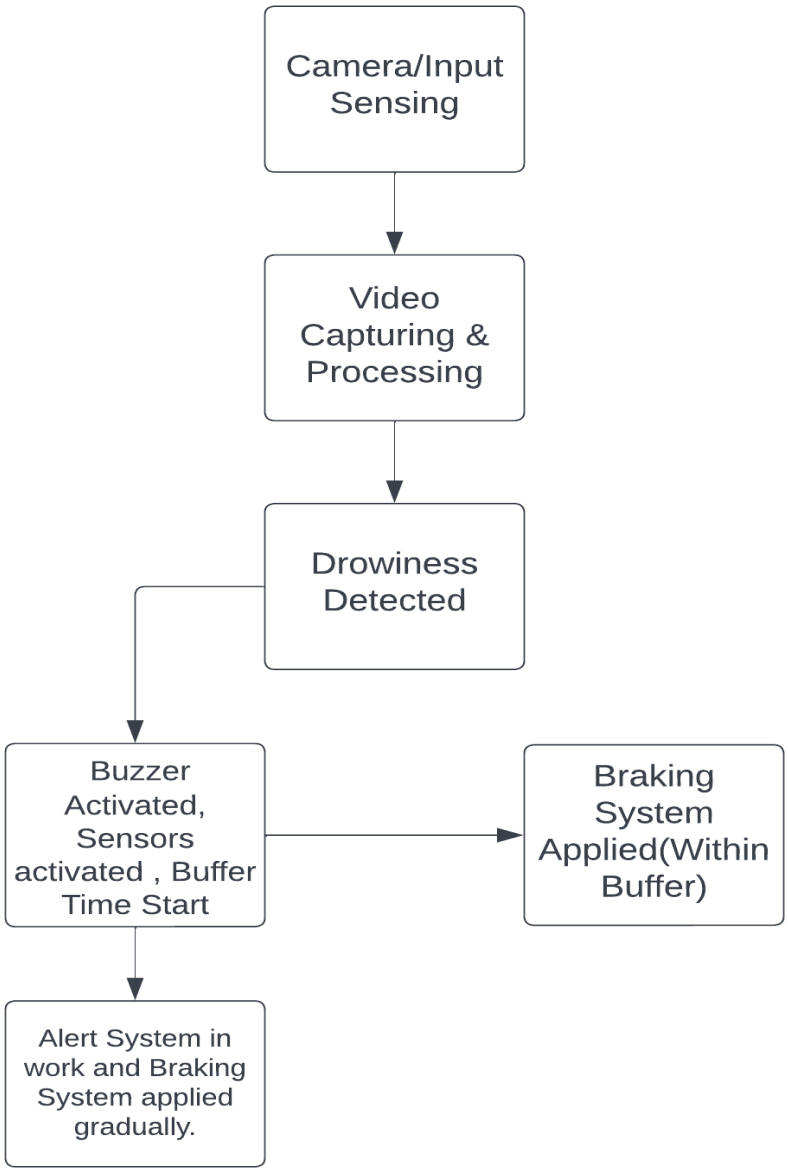


Figure 5: Block Diagram

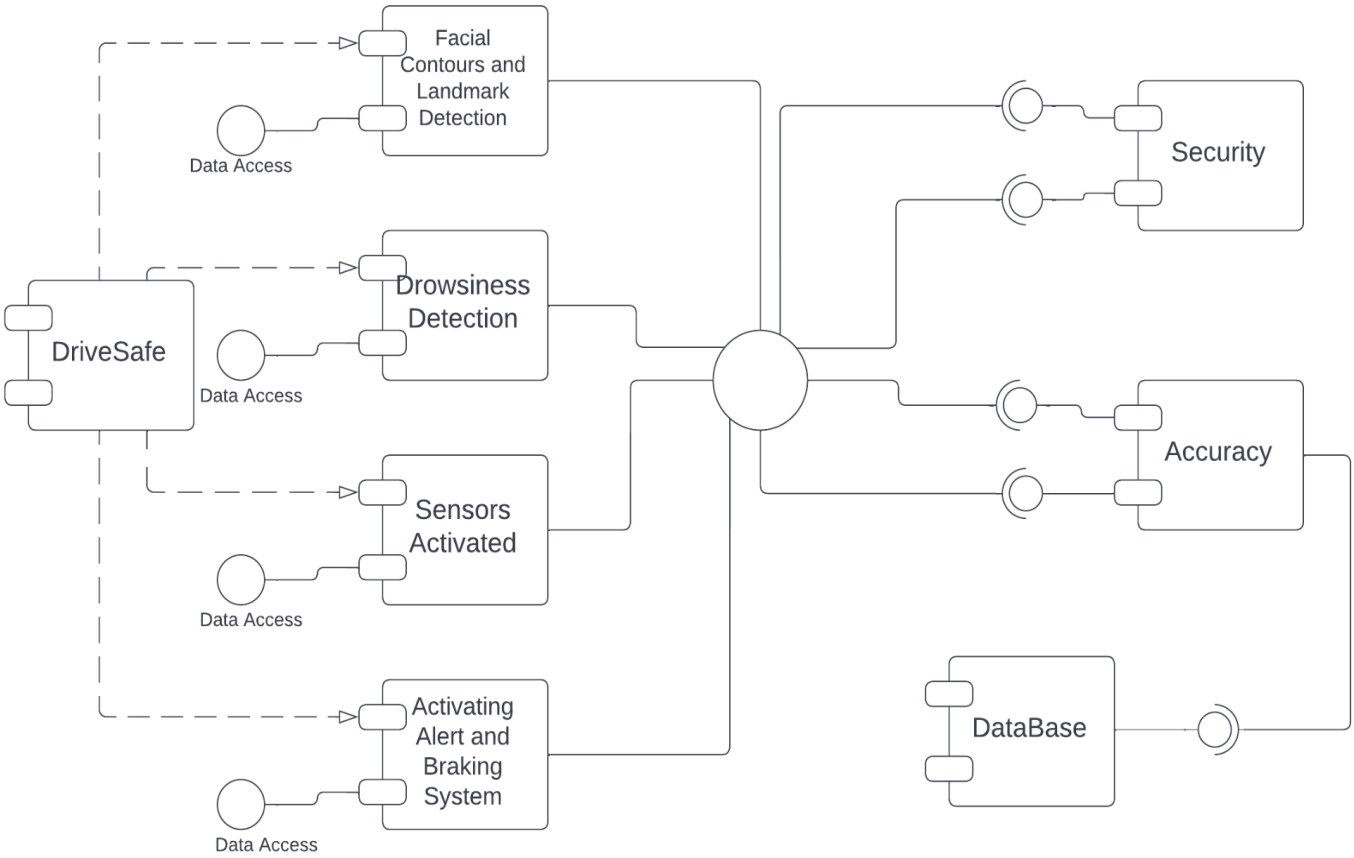


Figure 6: Component Diagram

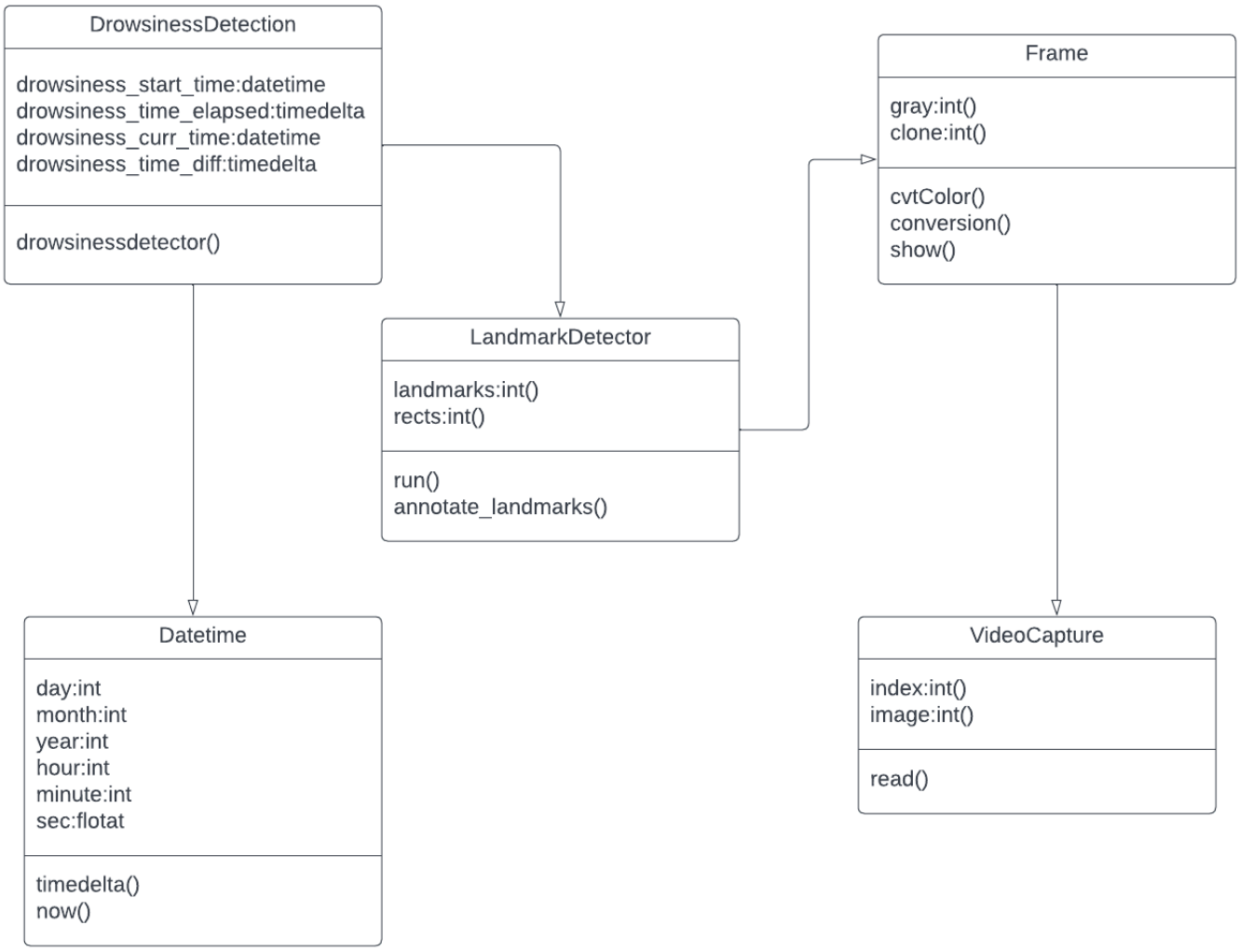


Figure 7: Class Diagram

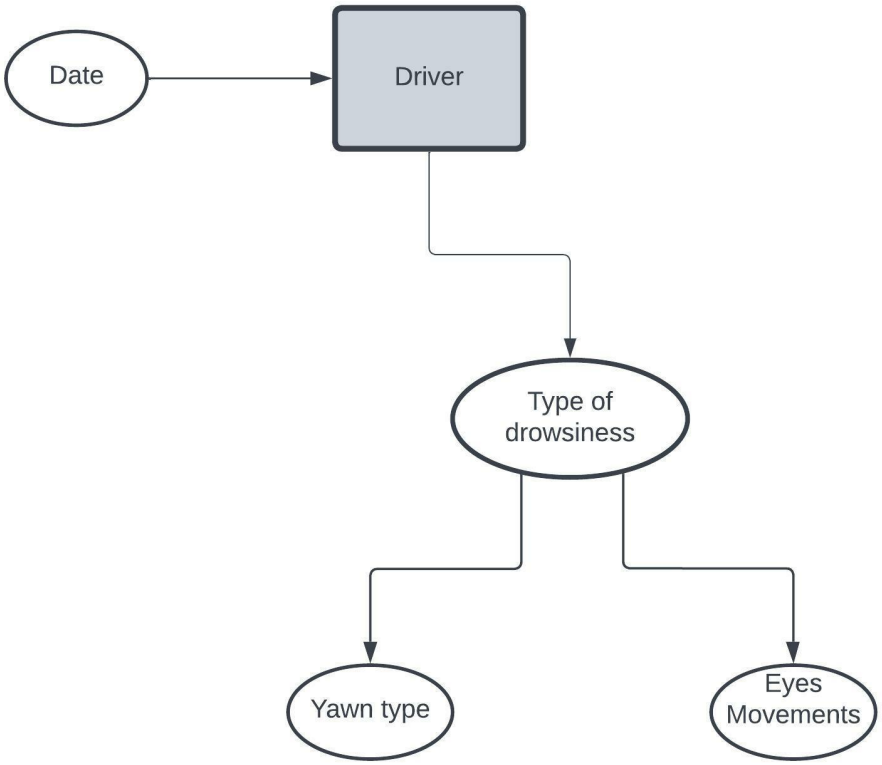


Figure 8: ER Diagram

4.2 Design Level Diagram

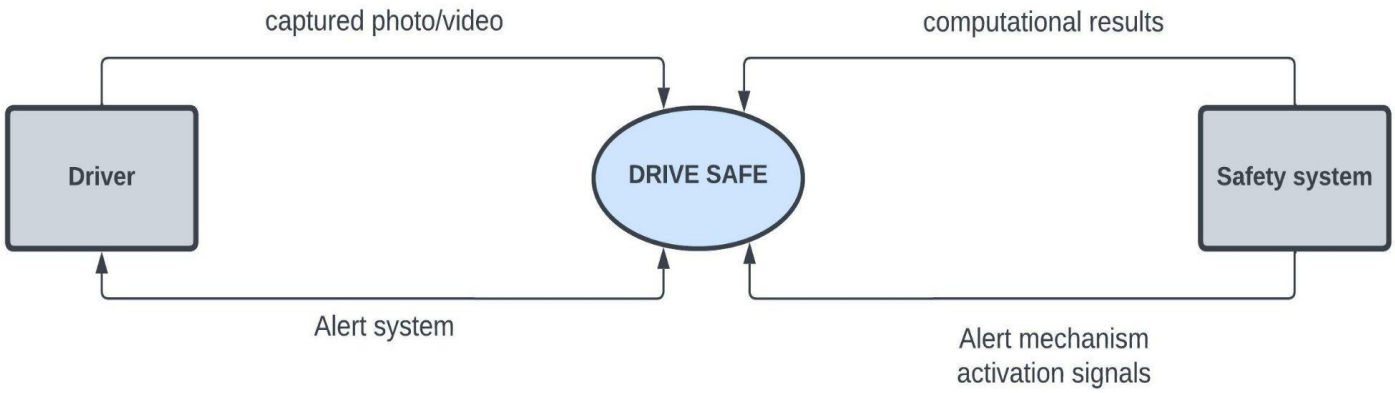


Figure 9: Level 0 DFD

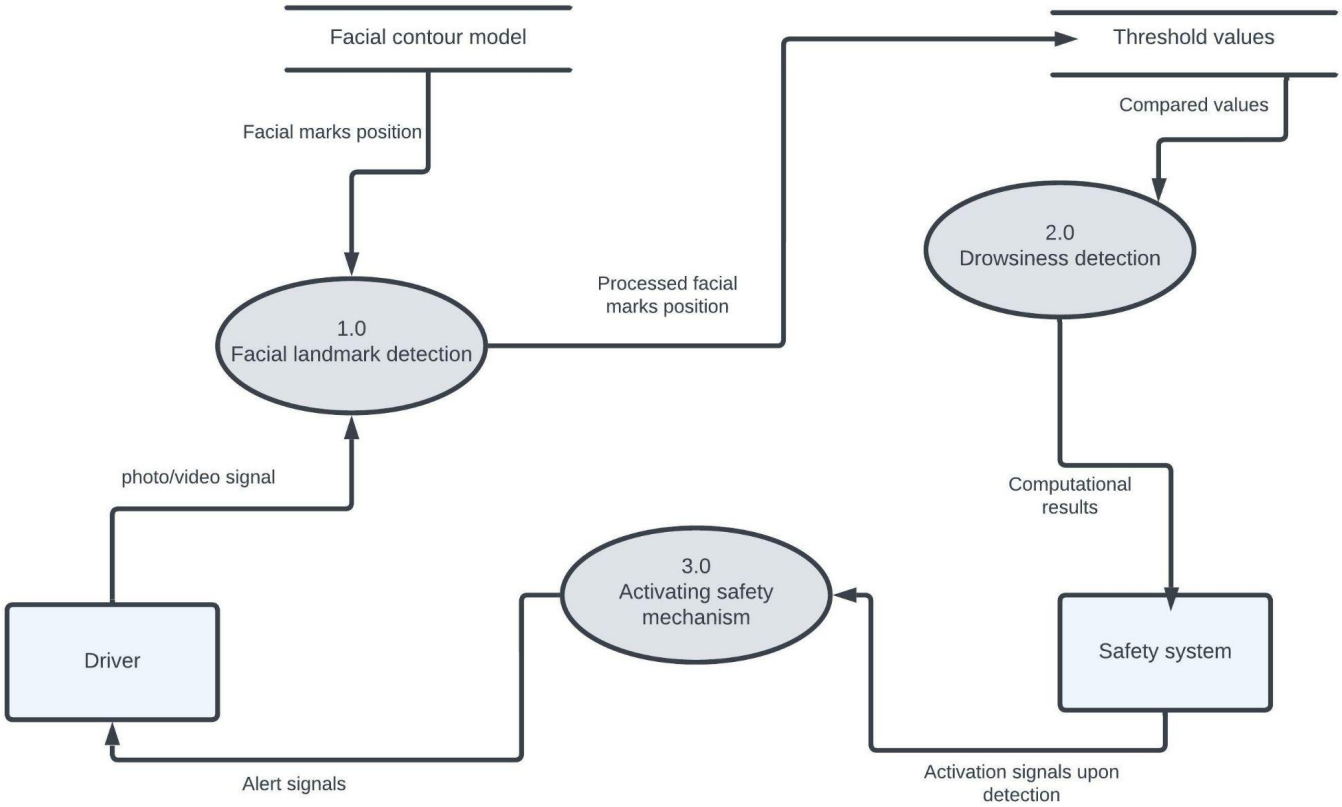


Figure 10: Level 1 DFD

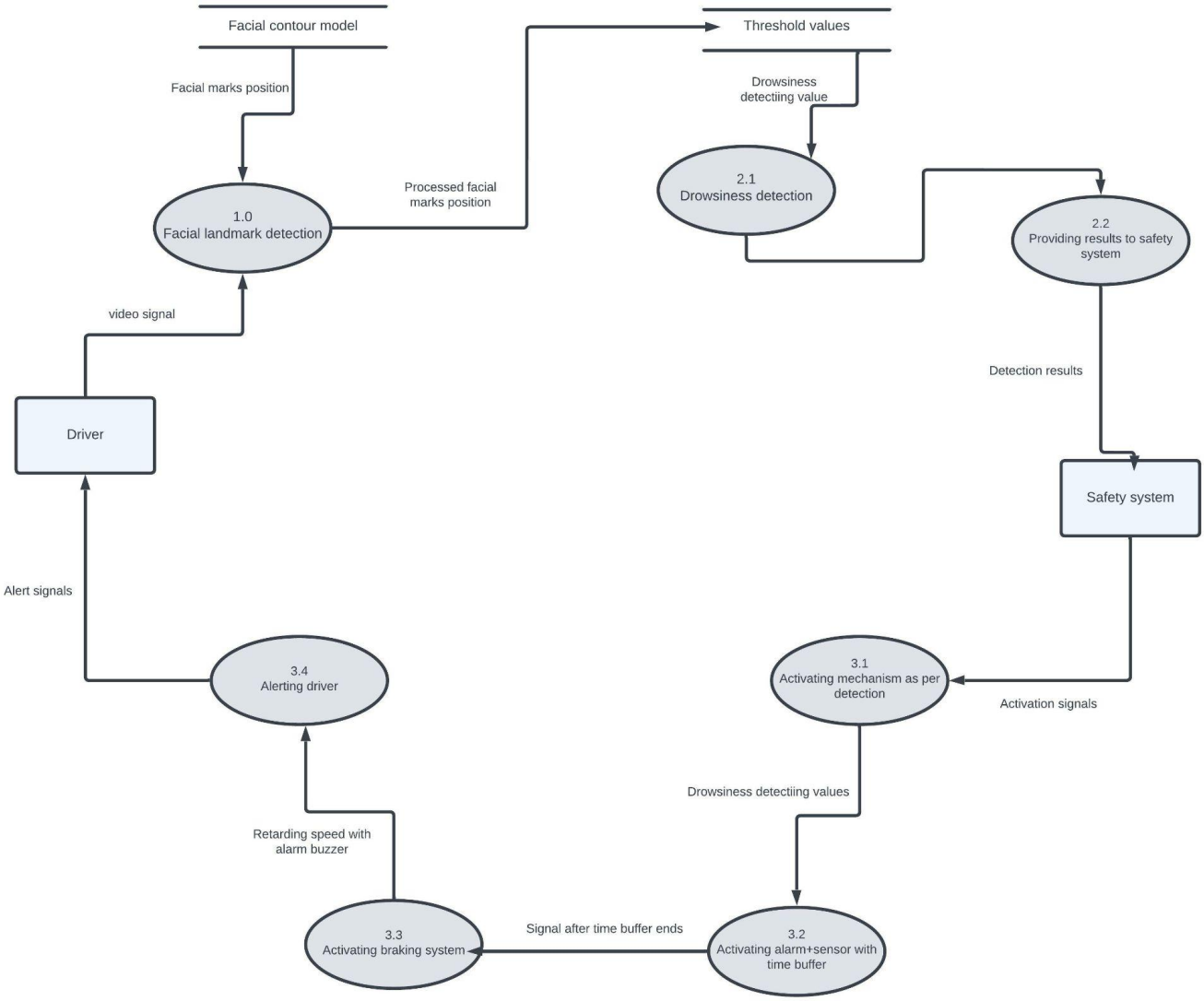


Figure 11: Level 2 DFD

4.3 User Interface Diagram

Use Case Diagram 1

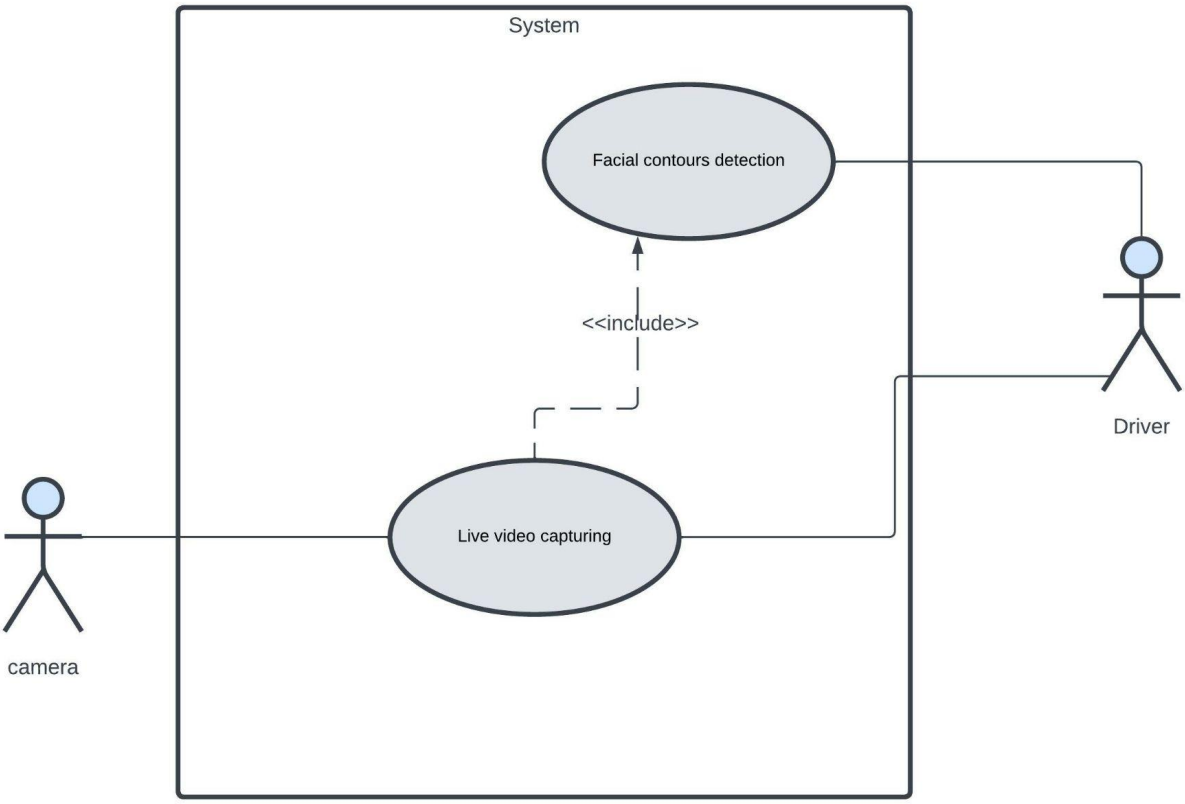


Figure 12: Use case 1

1.	Use Case Title	Video Capturing
2.	Abbreviated title	Video Capturing
3.	Use Case ID	1
4.	Actors	Camera, Driver
5.	Description	The real time video of the driver is captured and processed for the facial landmarks detection using the dlib libraries
5.1	Pre-Conditions	Power should be available to the camera
5.2	Task Sequence	1. Video is Captured 2. Video is Processed and facial contours are detected
5.3	Post Condition	The Facial contours detected are used as the input for the drowsiness detection system
6.	Modification History	24-06-2022
7.	Author	Jagjeevan Singh, Sai Sharan Sakilam, Saksham Chauhan

Table 6: Use case 1 Template

The flow of events:-

1. **Basic Flow**

   Camera:

   1. (SA): Capturing Drivers Real-time Video
   2. (SA): Video Processing and Detecting Facial Contours

2. **Alternate Flow**

   Driver:

   1. (AA) The driver has turned off the power to the monitoring System

3. **Exceptional Flow**

   Camera:

   1. (SA): Blurry video is being captured due to the dust over the camera
   2. (SR): Any Machinery Failure

Use Case Diagram 2

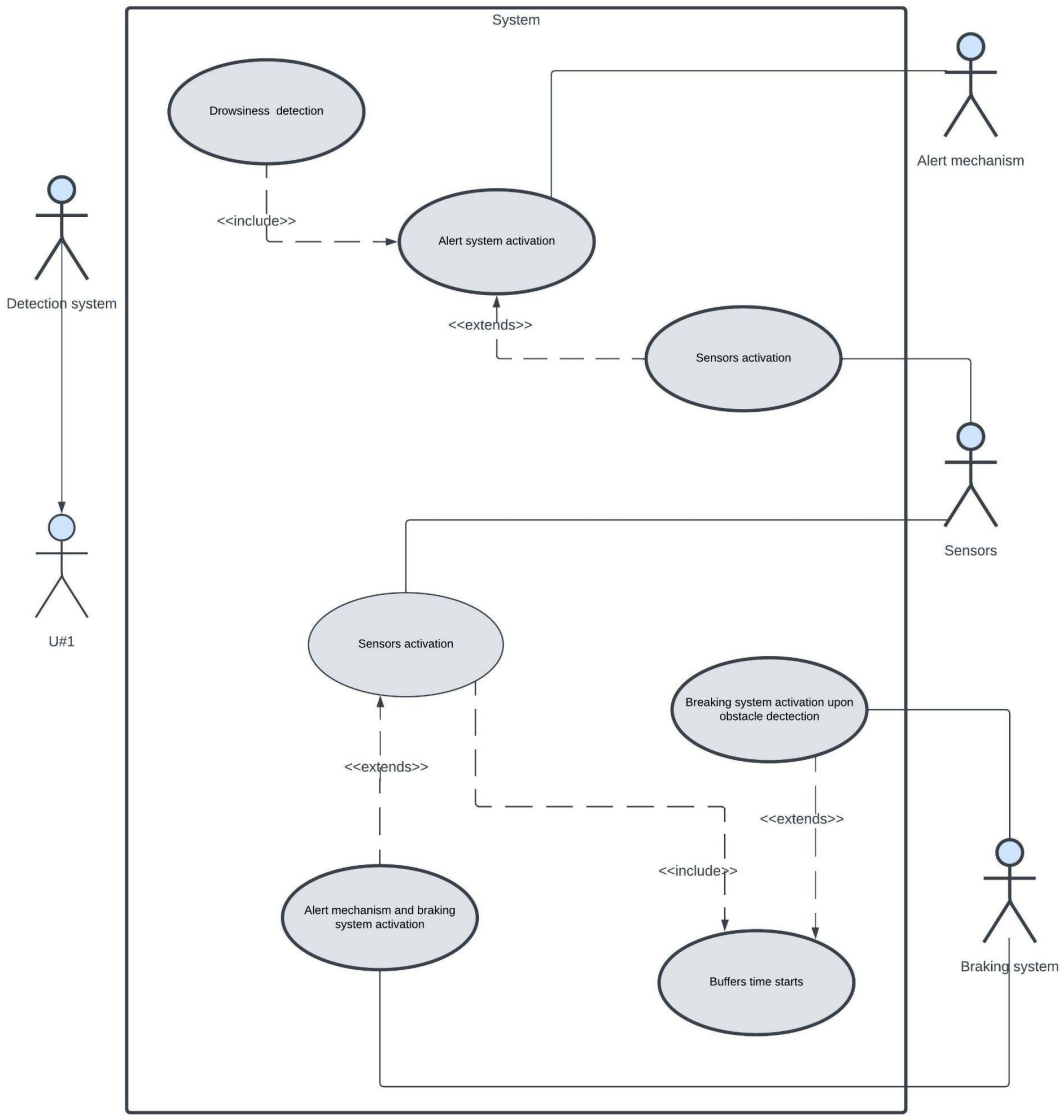


Figure 13: Use Case 2

1.	Use Case Title	Detection and Alert system
2.	Abbreviated title	Detection and Alert system
3.	Use Case ID	2
4.	Actors	Distraction System, Alert Mechanism, Sensors and Braking system
5.	Description	Given the detected facial contours, the Detection system will detect the driver's drowsiness and corresponding alert systems will be activated
5.1	Pre-Conditions	Availability of real-time video of driver
5.2	Task Sequence	Based upon Facial Contours, Firstly Detection System will detect the driver's drowsiness or distraction 1. Upon drowsiness detection, Alert Mechanism will be activated. 2. Buffer time will start 3. Buffer time on crossing the threshold value will lead to the activation of sensors and parallely braking system will be activated. 4. If any obstacle is detected within the buffer time the braking system will be applied. Once the buffer time is over and the driver does't get alert the braking system will be applied.
5.3	Post Condition	None
6.	Modification History	24-06-2022
7.	Author	Jagjeevan Singh, Sai Sharan Sakilam, Saksham Chauhan

Table 7: Use case 2 template

The flow of events:-

1. **Basic Flow**

   Camera:

   1. (SA): Detecting drowsiness
   2. (SR): Corresponding Alert mechanisms are performed

2. **Alternate Flow**

   Driver:

   1. (SR) Drowsiness was detected, followed by the sensor activation with the start of the Buffer timer
   2. (AA) Driver is alert within that Buffer time

3. **Exceptional Flow**

   Alert Mechanism:

   1. (SR) Any Hardware Failure along with breakage issue
   2. (AA) The driver is alert, but unfortunately, the driver meets with an accident due to other reasons.

4.4 Snapshots of Working Prototype

When the person is alert with no signs of drowsiness. The person is tagged as ALERT.

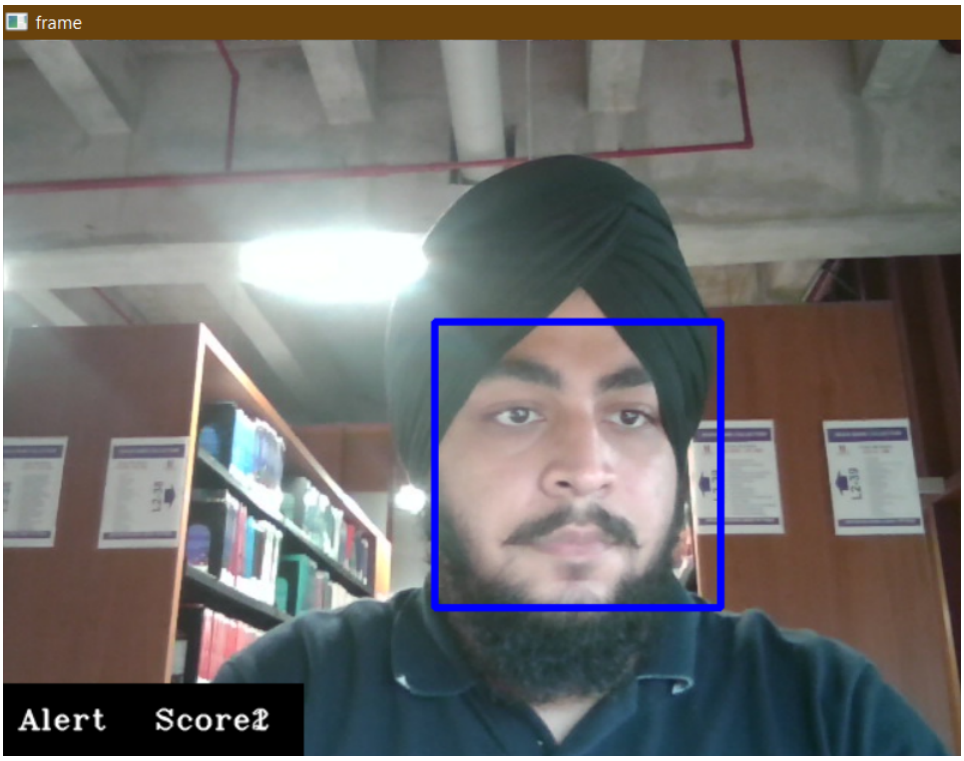


Figure 14: Alert Condition

When the person is dozing(Type of Drowsiness). Code successfully detects the drowsy condition and the person is tagged as DROWSY. Also the score is displayed along with i.e for how many seconds the person is dozing off. This score will act as a buffer timer.

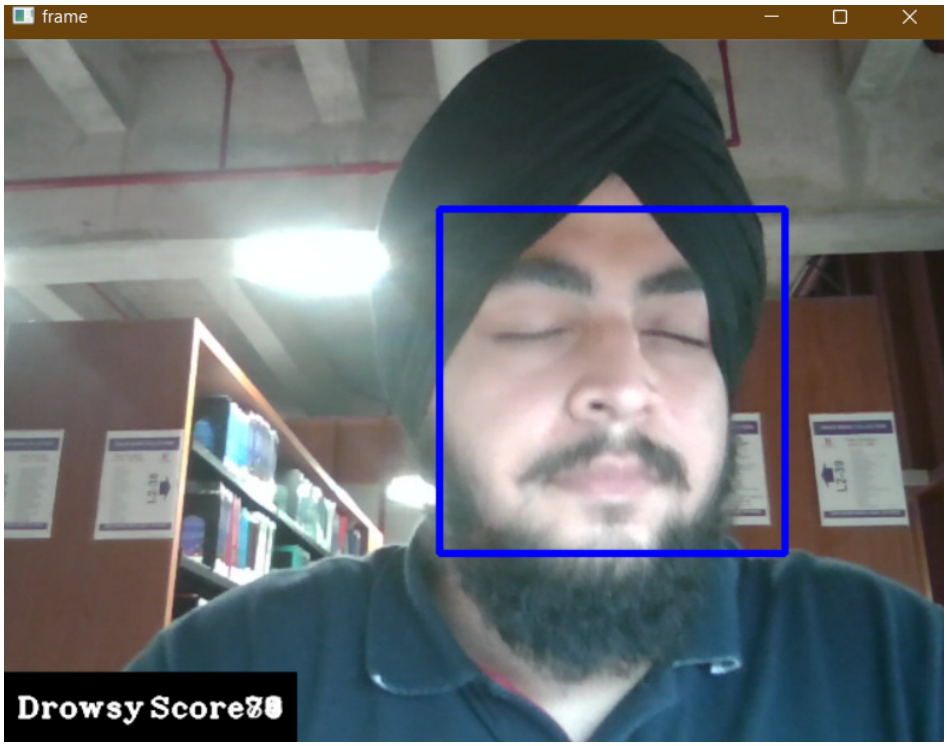


Figure 15: Drowsy Condition

5.1 Work Accomplished

- The project's need and scope were examined in order to make it more useful and to analyze the tasks required to complete it.
- All requirements were taken into consideration for improved project planning.
- A feasibility study was carried out in order to make the project more efficient and less time consuming by assisting in the decision-making process.
- Studying various articles and research papers provided us with ideas for areas where we can improve our project's reliability and robustness.
- An image enhancement approach is used to improve detection at night and in cloudy conditions.
- Facial landmarks are detected to detect the drowsiness, and further contours around the eyes, and mouth are analyzed.
- An algorithm generates various warnings based on the type and intensity of drowsiness to alert the driver.

5.2 Conclusion

An advanced safety system **'DRIVESAFE'** that utilizes image processing and image detection to detect drowsiness in order to ensure driver's safety has been proposed. We consider our product to be unique and one of a kind in the market for, The existing safety system had some flaws which have been solved in our solution and not only this but also many new features have been used in our solution to achieve more accuracy and have best possible efficiency in order to prevent driver's life.

5.3 Environmental, Economic and Social Benefits

Environmental Benefits:

The technology attempts to reduce the number of car collisions as well as vehicle damage that cannot be repaired and will wind up in a landfill for up to 80 years, obstructing space. Tree and plant damage along the roadway will also be reduced. The number of animals killed in highway collisions with speeding vehicles will be reduced as well. In a way, implementing a vehicle safety system would work towards the goal of sustainable development.

Economic Benefits

The system will leverage pre-existing technology in automobiles to assess surrounding traffic, such as sensors and wiring. The only additional cost would be for the camera to be installed and connected to the

hardware. By employing refurbished cameras, this cost can be reduced. There is no demand for any high-tech equipment. Once implemented, this approach has the potential to save up to 91 lakh Indian rupees per individual.

Social Benefits

A smart car safety system would be beneficial to society in a variety of ways. With fewer fatalities in car accidents, the risk of death or long-term bodily damage is lowered, as is the risk of emotional harm to the victim and their loved ones.

5.4 Future Work Plan

- ❖ Including Yawn detection corresponding to the eye and nose tracking.
- ❖ Making the buggy as our hardware prototype.
- ❖ Setting up the sensors on the buggy.
- ❖ Integration of the hardware and software components.
- ❖ Testing of the prototype and debugging the software part if needed.
- ❖ Once we are ready with our proposed solution we will try to incorporate this product in real market.

Activity	Month	Feb	Feb	Feb	Feb	March	March	March	March	April	April	April	April	May	May	May	May
	Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Problem identification	Plan																
Problem identification	Actual																
Literature survey	Plan																
Literature survey	Actual																
SRS	Plan																
SRS	Actual																
Programming	Plan																
Programming	Actual																
Hardware assembly	Plan																
Hardware assembly	Actual																
Integrating hardware and software	Plan																
Integrating hardware and software	Actual																
Debugging and testig	Plan																
Debugging and testig	Actual																
Report writing	Plan																
Report writing	Actual																

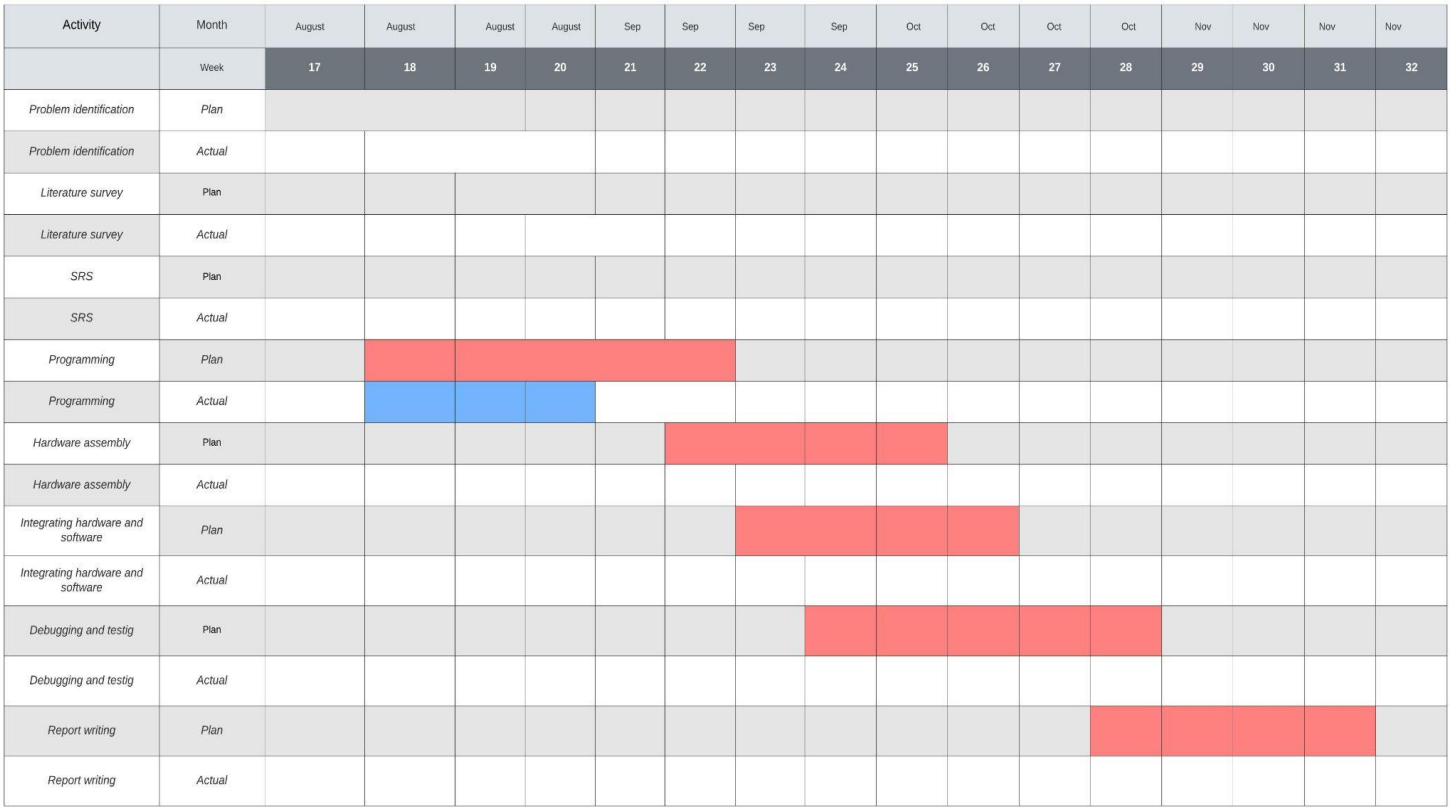


Figure 16: Future Work Plan

APPENDIX A: REFERENCES

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